

Show understanding of binary magnitudes and the difference between binary prefixes and decimal prefixes

9608_s20_qp_13_sc_7.b

QUESTION

(b) The finished logo is 160 pixels wide and 160 pixels high. The image has a colour depth of 3 bytes per pixel.

Calculate an estimate of the file size for the logo. Give your answer in kilobytes. Show your working.

Working
.....
.....

Answer KB

[3]

MARK SCHEME

7(b)	1 mark per bullet point	3
	160 * 160 (= 25600 pixels) 25600 * 3 (= 76800 bytes) 75 KB (divide by 1024) or 76.8 KB (divide by 1000)	

9608_s20_qp_11_sc_4.a

QUESTION

4 A digital camera takes a bitmap image. The image is 2000 pixels wide by 1000 pixels high with a colour depth of 24-bits.

(a) Calculate an estimate of the file size for the image. Give your answer in megabytes. Show your working.

Working
.....
.....
.....
.....

Answer MB

[3]

MARK SCHEME

4(a)	1 mark per bullet point • $2000 \times 1000 \times 24 = 48\,000\,000$ bits • $48\,000\,000 / 8 / 1024 / 1024$ • $= 6$ MB or 5.7 MB	3
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9608_w18_qp_13_sc_1.b.i

QUESTION

(b) (i) The designer takes a photograph to put on the poster. The photograph has a resolution of 50 000 pixels by 50 000 pixels. The colours are represented using 4 bytes per pixel.

Estimate the file size of the photograph in gigabytes. Show your working.

Working

.....

.....

.....

Estimated file size [4]

MARK SCHEME

1(b)(i)	1 mark per bullet point • Number of pixels: $50\,000 \times 50\,000 = 2\,500\,000\,000$ • Number of bytes: Number of pixels $(2\,500\,000\,000) \times \underline{4}$ // $10\,000\,000\,000$ // • Number of bits: Number of pixels $(2\,500\,000\,000) \times \underline{32}$ // $80\,000\,000\,000$ • Conversion to gigabytes • 10 GB / 9.3 GB	4
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QUESTION

- (b) The company takes a photograph of their office to put on the website. The photograph has a resolution of 1000 pixels by 1000 pixels. Two bytes per pixel are used to represent the colours.

- (i) Estimate the file size of the photograph in megabytes. Show your working.

Working

.....

.....

.....

Estimated file size

[4]

MARK SCHEME

1(b)(i)	1 mark per bullet point • Number of pixels: $1000 \times 1000 // 1\,000\,000$ • Number of bytes: Number of pixels ($1\,000\,000$) $\times 2 // 2\,000\,000 //$ Number of bits: Number of pixels ($1\,000\,000$) $\times 16 // 16\,000\,000$ • Conversion to <u>megabytes</u> • 2 (MB) // 1.91 (MB)	4
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QUESTION

- (c) The finished logo is 500 pixels by 1000 pixels and uses 35 different colours.

Estimate the file size for the logo. Give your answer in kilobytes. Show your working.

Working

.....

.....

.....

.....

Answer

[4]

MARK SCHEME

2(c)	1 mark per bullet Number of pixels $500 \times 1000 (= 500\,000)$ 35 colours require 6 bits per pixel Number of bytes $(500\,000 \times 6) / 8 = 3\,000\,000 / 8 (= 375\,000)$ = 375 Kb	4
------	--	----------

9608_s17_qp_13_sc_3.b.iii

QUESTION

- (iii) A second picture has width 2048 pixels and height 512 pixels. It is saved as a 256-colour image.

Calculate the file size in kilobytes.

Show your working.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

3(b)(iii)	<i>Working: Max two from:</i> Number of pixels is 2048×512 1 One pixel will be stored as one byte 1 Number of kilobytes = $(2048 \times 512) / 1024$ 1 <i>Answer: One mark:</i> Number of kilobytes = 1024 KB 1	3
-----------	---	----------

QUESTION

(iii) A second picture has width 8192 pixels and height 256 pixels. It is saved as a 256-colour bitmap.

Calculate the file size in kilobytes.

Show your working.

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.....[3]

MARK SCHEME

3(b)(iii)	<i>Working: Max two from:</i> Number of pixels is $8192 \cdot 256$ 1 One pixel will be stored as one byte 1 Number of kilobytes = $(8192 \cdot 256) / 1024$ 1 <i>Answer: One mark:</i> Number of kilobytes = 2048 KB 1	3
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QUESTION

(iii) A second picture has width 2048 pixels and height 512 pixels. It is saved as a 256-colour image.

Calculate the file size in kilobytes.

Show your working.

.....

.....

.....

.....

.....

.....[3]

MARK SCHEME

3(b)(iii)	Working: Max two from: Number of pixels is $2048 \cdot 512$ 1 One pixel will be stored as one byte 1 Number of kilobytes = $(2048 \cdot 512) / 1024$ 1 Answer: One mark: Number of kilobytes = 1024 KB 1	3
-----------	--	----------

9608_w15_qp_13_sc_8.b.i

QUESTION

(b) (i) A black and white image is 512 pixels by 256 pixels.

Calculate the file size of this image in kilobytes (KB) (1 KB = 1024 bytes).
Show your working.

.....

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.....

.....

.....[2]

MARK SCHEME

(b) (i) $\frac{512 \times 256}{8 \times 1024} = 16 \text{ KB}$

1 mark for numerator + 1 mark for denominator

[2]

Show understanding of different number systems

9608_s21_qp_13_sc_3.c.iii

QUESTION

- (iii) The contents of memory address 190 represent a two's complement binary integer.

Address

190

1	1	0	1	1	0	0	1
---	---	---	---	---	---	---	---

Convert the value in memory address 190 into denary.

..... [1]

MARK SCHEME

3(c)(iii)	–39	1
-----------	-----	---

9608_s21_qp_11_sc_3.c.ii

QUESTION

- (ii) State the **maximum** number of op codes that can be represented using eight bits.

..... [1]

MARK SCHEME

3(c)(ii)	256	1
----------	-----	---

9608_s20_qp_12_sc_5.b.iii

QUESTION

- (iii) State why the denary value for the op code ADD cannot be represented in 8-bit two's complement form. Justify your answer.

.....
.....
.....
..... [2]

MARK SCHEME

5(b)(iii)	1 mark per bullet point to max 2 - The maximum range for an 8-bit two's complement binary number is –128 to +127 ... 200 is outside of the maximum range	2
-----------	--	----------

9608_w19_qp_13_sc_4.a

QUESTION

4 (a) Convert the unsigned binary number 0101 1111 1100 into denary. [1]

MARK SCHEME

4(a)	1532	1
------	------	----------

9608_w19_qp_13_sc_4.b

QUESTION

(b) Convert the denary number –239 into 12-bit two's complement. [1]

MARK SCHEME

4(b)	1111 0001 0001	1
------	----------------	----------

9608_w19_qp_13_sc_4.c

QUESTION

(c) Convert the two's complement number 0110 0101 into denary. [1]

MARK SCHEME

4(c)	101	1
------	-----	----------

9608_w19_qp_13_sc_4.d

QUESTION

(d) Convert the Binary Coded Decimal (BCD) value 0110 0101 into denary.

..... [1]

MARK SCHEME

4(d)	65	1
------	----	---

9608_w19_qp_13_sc_4.e

QUESTION

(e) Convert the denary number 222 into hexadecimal.

..... [1]

MARK SCHEME

4(e)	DE	1
------	----	---

9608_w18_qp_13_sc_2.b.iii

QUESTION

(iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

2(b)(iii)	-14	1
-----------	-----	---

9608_s18_qp_13_sc_3.c.iii

QUESTION

(iii) The current contents of register H represent a two's complement binary integer.

Convert the value in register H into denary.

.....[1]

MARK SCHEME

3(c)(iii)	– 63	1
-----------	------	---

9608_s18_qp_13_sc_3.c.iv

QUESTION

(iv) State why register H does not currently contain a Binary Coded Decimal (BCD).

.....

.....[1]

MARK SCHEME

3(c)(iv)	The <u>first 4 bits / first nibble</u> (would give 12 which) is <u>> 9 / 2 digits</u> (which is not valid for BCD)	1
----------	---	---

9608_s18_qp_12_sc_4.a.iii

QUESTION

(iii) The current contents of the Accumulator represent a two's complement binary integer.

Convert the value in the Accumulator into denary.

.....[1]

MARK SCHEME

4(a)(iii)	–37	1
-----------	-----	---

9608_s18_qp_11_sc_8.c.ii

QUESTION

(ii) The current contents of register X represent a Binary Coded Decimal.

Convert the value in X into denary.

.....[1]

MARK SCHEME

8(c)(ii)	87	1
----------	----	---

9608_s18_qp_11_sc_8.c.iii

QUESTION

(iii) The current contents of register X stores a two's complement binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

8(c)(iii)	−121	1
-----------	------	---

9608_w17_qp_13_sc_1.a.i

QUESTION

1 (a) Each of the following bytes represents an integer in two's complement form.

State the denary value.

(i) 0111 0111 Denary [1]

MARK SCHEME

1(a)(i)	119	1
---------	-----	---

9608_w17_qp_13_sc_1.a.ii

QUESTION

(ii) 1000 1000

Denary

[1]

MARK SCHEME

1(a)(ii)

–120

1

9608_w17_qp_13_sc_1.a.iii

QUESTION

(iii) Express the following integer in two's complement form.

–17

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

1(a)(iii)

1	1	1	0	1	1	1	1
---	---	---	---	---	---	---	---

1

9608_w17_qp_13_sc_1.a.iv

QUESTION

(iv) State in denary, the range of integer values that it is possible to represent in two's complement integers using a single byte.

Lowest value

Highest value

[1]

MARK SCHEME

1(a)(iv)

Lowest value: –128
Highest value: +127

1

9608_w17_qp_13_sc_1.b.i

QUESTION

(b) (i) Convert the following denary integer into Binary Coded Decimal (BCD).

653

.....[1]

MARK SCHEME

1(b)(i)	0110 0101 0011	1
---------	----------------	---

9608_w17_qp_13_sc_1.b.ii

QUESTION

(ii) A 3-digit BCD representation has been incorrectly copied. It is shown as:

0	1	0	0	1	1	1	0	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---

State how you can recognise that this is not a valid BCD representation.

.....

.....[1]

MARK SCHEME

1(b)(ii)	The second block of four binary digits represents a digit larger than 9 // 14	1
----------	---	---

9608_w16_qp_13_sc_3.a.ii

QUESTION

(ii) Convert the denary integer –46 to an 8-bit two’s complement form.

.....

.....[1]

MARK SCHEME

(ii)	1 1 0 1 0 0 1 0	[1]
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QUESTION

(b) Binary Coded Decimal (BCD) is another way of representing numbers.

- (i) Describe how denary integers larger than 9 can be converted into BCD.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(b) (i) **One mark** for the explanation and **one mark** for the example

- Each denary digit is written as a 4-bit binary number
- Example: 46 = 0100 0110

[2]

QUESTION

- (ii) Describe how an 8-bit BCD representation can be converted into a denary integer.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(ii) **One mark** for the explanation and **one mark** for the example

- Binary number is split up into groups of 4 bits (starting from the right)
// Each group of 4 bits is converted to a denary digit
- Example: 0011 0111 = 37

[2]

QUESTION

- 4 Hexadecimal, Binary Coded Decimal (BCD) and binary values are shown below.

Draw a line to link each value to its correct denary value.

Hexadecimal, BCD, binary**Denary**

Hexadecimal:

3A

93

-65

BCD representation:

0100 1001

58

-63

Binary integer:

01011101

73

Two's complement
binary integer:

11000001

49

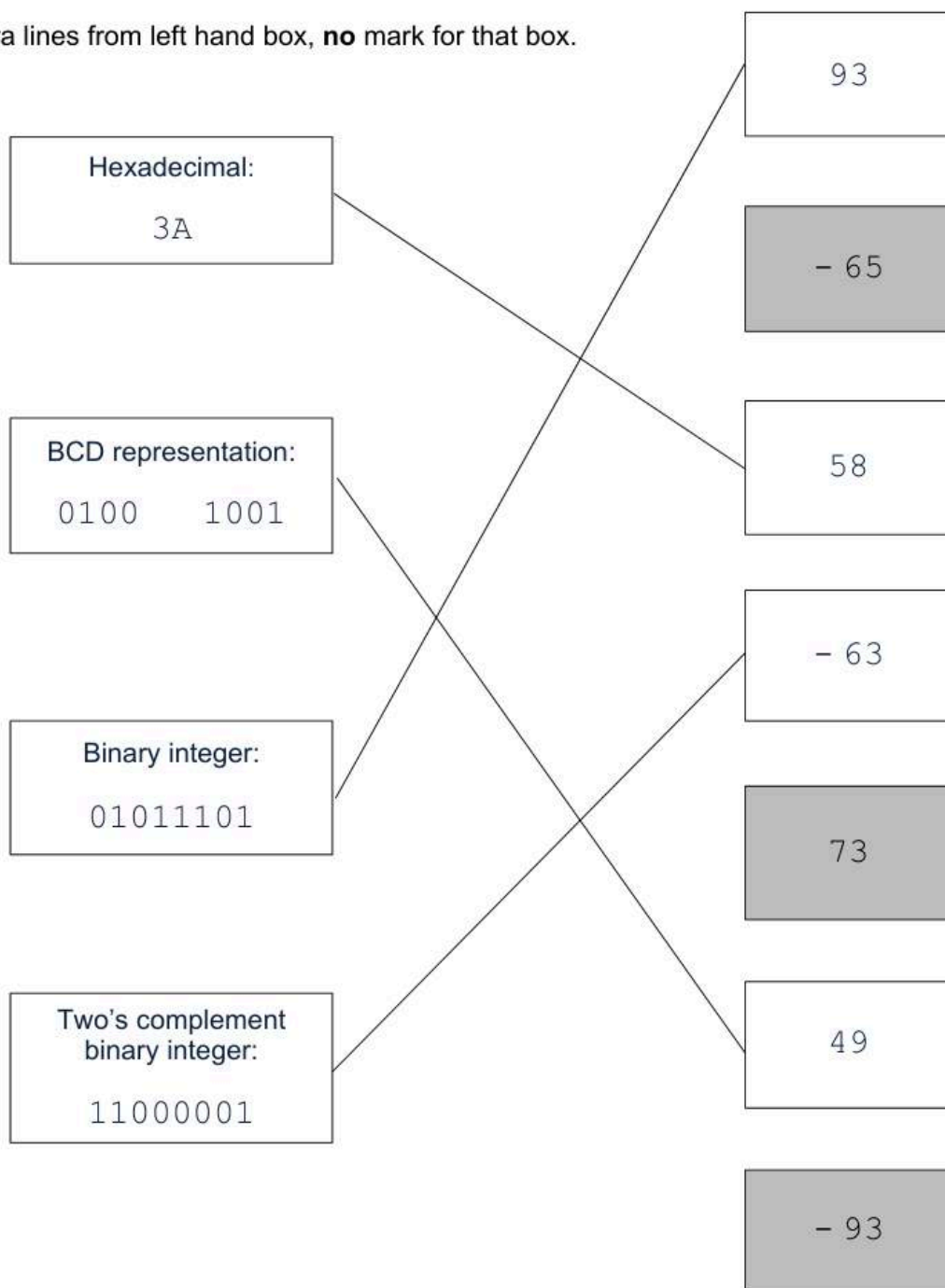
-93

[4]

MARK SCHEME

4 **ONE** mark for each correct line.

Extra lines from left hand box, **no** mark for that box.



[4]

QUESTION

(b) Binary Coded Decimal (BCD) is another way of representing numbers.

- (i) Describe how denary integers larger than 9 can be converted into BCD.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(b) (i) **One mark** for the explanation and **one mark** for the example

- Each denary digit is written as a 4-bit binary number
- Example: 46 = 0100 0110

[2]

QUESTION

- (ii) Describe how an 8-bit BCD representation can be converted into a denary integer.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(ii) **One mark** for the explanation and **one mark** for the example

- Binary number is split up into groups of 4 bits (starting from the right)
// Each group of 4 bits is converted to a denary digit
- Example: 0011 0111 = 37

[2]

9608_s16_qp_13_sc_2.b

QUESTION

(b) Convert the following denary number into Binary Coded Decimal (BCD).

82

.....[1]

MARK SCHEME

(b) 1000 0010

[1]

9608_s16_qp_13_sc_2.c

QUESTION

(c) Convert the following two's complement integer number into denary.

11001011

.....
.....
.....[2]

MARK SCHEME

(c) -53

[2]

One mark for '53' and **one** mark for '-'

9608_s16_qp_12_sc_2.b

QUESTION

(b) Convert the following Binary Coded Decimal (BCD) number into denary.

10000011

.....[1]

MARK SCHEME

(b) 83 [1]

9608_s16_qp_12_sc_2.c

QUESTION

(c) Convert the following denary integer into 8-bit two's complement.

-102

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(c) 10011010 [2]

Marks allocated as follows:

1 mark for the most significant bit
1 mark for the remaining 7 bits

9608_s16_qp_11_sc_2.b

QUESTION

(b) Convert the following Binary Coded Decimal (BCD) number into denary.

10000011

.....[1]

MARK SCHEME

(b) 83 [1]

QUESTION

(c) Convert the following denary integer into 8-bit two's complement.

-102

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(c) 10011010

[2]

Marks allocated as follows:

1 mark for the most significant bit

1 mark for the remaining 7 bits

QUESTION

1 (a) (i) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

124

--	--	--	--	--	--	--	--

-77

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

1 (a) (i)

124	0	1	1	1	1	1	0	0
-77	1	0	1	1	0	0	1	1

[2]

9608_s15_qp_13_sc_1.a.ii

QUESTION

(ii) Convert the two numbers in **part (a) (i)** into hexadecimal.

124

-77 [2]

MARK SCHEME

(ii) 124: 7 C

-77: B 3

[2]

9608_s15_qp_13_sc_1.b.i

QUESTION

(b) Binary Coded Decimal (BCD) is another way of representing numbers.

(i) Write the number 359 in BCD form.

..... [1]

MARK SCHEME

(b) (i) 0 0 1 1 0 1 0 1 1 0 0 1

[1]

9608_s15_qp_12_sc_1.ii

QUESTION

(ii) Convert the following denary number into BCD format.

9 7

..... [1]

MARK SCHEME

(ii) 1 0 0 1 0 1 1 1

[1]

9608_s15_qp_12_sc_1.iii

QUESTION

- (iii) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

114

--	--	--	--	--	--	--	--

- 93

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(iii)

114

0	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---

- 93

1	0	1	0	0	0	1	1
---	---	---	---	---	---	---	---

[2]

9608_s15_qp_11_sc_1.iii

QUESTION

- (iii) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

114

--	--	--	--	--	--	--	--

- 93

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(iii)

114

0	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---

– 93

1	0	1	0	0	0	1	1
---	---	---	---	---	---	---	---

[2]

Convert an integer value from one number base/representation to another

9608_s21_qp_13_sc_3.c.i

QUESTION

(c) Each instruction in the assembly language program is encoded in 16 bits (8-bit op code followed by an 8-bit operand).

(i) The instruction `LDX 234` has the operand 234.

Convert the operand 234 into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

3(c)(i)	1110 1010	1
---------	-----------	---

9608_s21_qp_13_sc_3.c.ii

QUESTION

(ii) Convert the denary value 234 into hexadecimal.

..... [1]

MARK SCHEME

3(c)(ii)	EA	1
----------	----	---

9608_s21_qp_13_sc_3.c.iii

QUESTION

- (iii) The contents of memory address 190 represent a two's complement binary integer.

Address

190	1	1	0	1	1	0	0	1
-----	---	---	---	---	---	---	---	---

Convert the value in memory address 190 into denary.

..... [1]

MARK SCHEME

3(c)(iii)	–39	1
-----------	-----	---

9608_s21_qp_12_sc_3.c.i

QUESTION

- (c) Each instruction in the assembly language program is encoded in 16 bits (8-bit op code followed by an 8-bit operand).

- (i) The instruction `JPE 59` has the operand 59.

Convert the operand 59 into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

3(c)(i)	0011 1011	1
---------	-----------	---

9608_s21_qp_12_sc_3.c.ii

QUESTION

- (ii) Convert the denary value 59 into hexadecimal.

..... [1]

MARK SCHEME

3(c)(ii)	3B	1
----------	----	---

9608_s21_qp_11_sc_3.c.i

QUESTION

(c) Each instruction in the assembly language program is encoded in 16 bits (8-bit op code followed by an 8-bit operand).

(i) The instruction `CMP 201` has the operand 201.

Convert the operand 201 into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

3(c)(i)	1100 1001	1
---------	-----------	---

9608_s20_qp_13_sc_7.d

QUESTION

(d) The hexadecimal colour value of the background of Xiaoming's website is:

913C8E

Complete the following table by converting each hexadecimal value to denary value.

	Red	Green	Blue
Hexadecimal value	91	3C	8E
Denary value			

[2]

MARK SCHEME

7(d)	1 mark for 1 or 2 correct answers 2 marks for 3 correct answers	2												
	<table><tr><td></td><td>Red</td><td>Green</td><td>Blue</td></tr><tr><td>Hexadecimal value</td><td>91</td><td>3C</td><td>8E</td></tr><tr><td>Denary value</td><td>145</td><td>60</td><td>142</td></tr></table>		Red	Green	Blue	Hexadecimal value	91	3C	8E	Denary value	145	60	142	
	Red	Green	Blue											
Hexadecimal value	91	3C	8E											
Denary value	145	60	142											

9608_s20_qp_12_sc_5.b.i

QUESTION

(b) The assembler translates assembly code into machine code.

The table shows the denary values for three assembler op codes.

Op code	Denary value
LDD	194
ADD	200
STO	205

(i) Convert the denary value for the op code LDD into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

5(b)(i)	11000010	1
---------	----------	---

9608_s20_qp_12_sc_5.b.ii

QUESTION

(ii) Convert the denary value for the op code STO into hexadecimal.

..... [1]

MARK SCHEME

5(b)(ii)	CD	1
----------	----	---

QUESTION

- (c) The digital camera allows a user to add text to an image. The text is encoded as ASCII values.

The table shows the ASCII denary values for five characters.

Character	ASCII denary value
a	97
b	98
c	99
d	100
e	101

- (i) Give the 8-bit binary value for the ASCII character 'b'.

.....
 [1]

MARK SCHEME

4(c)(i)	0110 0010	1
---------	-----------	---

QUESTION

- (ii) Complete the table by writing the ASCII denary value for the character 't' **and** its hexadecimal equivalent.

Character	t
ASCII denary value	
Hexadecimal value	

[2]

MARK SCHEME

4(c)(ii)	1 mark for each correct line	2						
	<table><tr><td>Character</td><td>t</td></tr><tr><td>ASCII denary value</td><td>116</td></tr><tr><td>Hexadecimal value</td><td>74</td></tr></table>	Character	t	ASCII denary value	116	Hexadecimal value	74	
Character	t							
ASCII denary value	116							
Hexadecimal value	74							

9608_w19_qp_13_sc_4.a

QUESTION

4 (a) Convert the unsigned binary number 0101 1111 1100 into denary.

..... [1]

MARK SCHEME

4(a)	1532	1
------	------	---

9608_w19_qp_13_sc_4.b

QUESTION

(b) Convert the denary number –239 into 12-bit two's complement.

..... [1]

MARK SCHEME

4(b)	1111 0001 0001	1
------	----------------	---

9608_w19_qp_13_sc_4.c

QUESTION

(c) Convert the two's complement number 0110 0101 into denary.

..... [1]

MARK SCHEME

4(c)	101	1
------	-----	---

9608_w19_qp_13_sc_4.d

QUESTION

(d) Convert the Binary Coded Decimal (BCD) value 0110 0101 into denary.

..... [1]

MARK SCHEME

4(d)	65	1
------	----	---

9608_w19_qp_13_sc_4.e

QUESTION

(e) Convert the denary number 222 into hexadecimal.

..... [1]

MARK SCHEME

4(e)	DE	1
------	----	---

9608_w19_qp_11_sc_5.a.ii

QUESTION

(ii) One binary colour is represented by 0100 1110

Convert the unsigned binary number 0100 1110 into denary.

..... [1]

MARK SCHEME

5(a)(ii)	78	1
----------	----	---

9608_w19_qp_11_sc_5.b

QUESTION

(b) Convert the denary number –194 into 12-bit two's complement.

..... [1]

MARK SCHEME

5(b)	1111 0011 1110	1
------	----------------	---

9608_w19_qp_11_sc_5.c.i

QUESTION

(c) (i) Convert the Binary Coded Decimal (BCD) value 0110 1001 into denary.

..... [1]

MARK SCHEME

5(c)(i)	69	1
---------	----	---

9608_s19_qp_12_sc_3.d.i

QUESTION

(d) The ASCII character code for 'A' is 65 in denary.

(i) Convert the denary ASCII character code for 'A' into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

3(d)(i)	1 mark for correct answer 0100 0001	1
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9608_s19_qp_12_sc_3.d.ii

QUESTION

(ii) Convert the denary ASCII character code for 'A' into hexadecimal.

..... [1]

MARK SCHEME

3(d)(ii)	1 mark for correct answer 41	1
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9608_w18_qp_13_sc_2.b.i

QUESTION

(b) The current contents of a general purpose register (X) are:

X	1	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---	---

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

2(b)(i)	242	1
---------	-----	---

9608_w18_qp_13_sc_2.b.ii

QUESTION

(ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

.....[1]

MARK SCHEME

2(b)(ii)	F2	1
----------	----	---

9608_w18_qp_13_sc_2.b.iii

QUESTION

(iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

2(b)(iii)	-14	1
-----------	-----	---

9608_w18_qp_12_sc_3.c.i

QUESTION

(c) The current contents of a general purpose register (X) are:

X	1	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---	---

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

..... [1]

MARK SCHEME

3(c)(i)	193	1
---------	-----	---

9608_w18_qp_12_sc_3.c.ii

QUESTION

(ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

..... [1]

MARK SCHEME

3(c)(ii)	C1	1
----------	----	---

9608_w18_qp_12_sc_3.c.iii

QUESTION

(iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

..... [1]

MARK SCHEME

3(c)(iii)	−63	1
-----------	-----	---

9608_w18_qp_11_sc_4.c.i

QUESTION

(c) The current contents of a general purpose register (X) are:

X	1	0	1	1	1	0	1	0
---	---	---	---	---	---	---	---	---

(i) The contents of X represent an unsigned binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

4(c)(i)	186	1
---------	-----	---

9608_w18_qp_11_sc_4.c.ii

QUESTION

(ii) The contents of X represent an unsigned binary integer.

Convert the value in X into hexadecimal.

.....[1]

MARK SCHEME

4(c)(ii)	BA	1
----------	----	---

9608_w18_qp_11_sc_4.c.iii

QUESTION

(iii) The contents of X represent a two's complement binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

4(c)(iii)	-70	1
-----------	-----	---

9608_s18_qp_13_sc_3.c.i

QUESTION

(c) H is a register. The current contents of H are:

1	1	0	0	0	0	0	1
---	---	---	---	---	---	---	---

The current contents of register H represent an unsigned binary integer.

(i) Convert the value in register H into denary.

.....[1]

MARK SCHEME

3(c)(i)	193	1
---------	-----	---

9608_s18_qp_13_sc_3.c.ii

QUESTION

(ii) Convert the value in register H into hexadecimal.

.....[1]

MARK SCHEME

3(c)(ii)	C1	1
----------	----	---

9608_s18_qp_13_sc_3.c.iii

QUESTION

(iii) The current contents of register H represent a two's complement binary integer.

Convert the value in register H into denary.

.....[1]

MARK SCHEME

3(c)(iii)	– 63	1
-----------	------	---

9608_s18_qp_12_sc_4.a.i

QUESTION

- 4 (a) The Accumulator is a register. The current contents of the Accumulator are:

1	1	0	1	1	0	1	1
---	---	---	---	---	---	---	---

The current contents of the Accumulator represent an unsigned binary integer.

- (i) Convert the value in the Accumulator into denary.

.....[1]

MARK SCHEME

4(a)(i)	219	1
---------	-----	---

9608_s18_qp_12_sc_4.a.ii

QUESTION

- (ii) Convert the value in the Accumulator into hexadecimal.

.....[1]

MARK SCHEME

4(a)(ii)	DB	1
----------	----	---

9608_s18_qp_12_sc_4.a.iii

QUESTION

- (iii) The current contents of the Accumulator represent a two's complement binary integer.

Convert the value in the Accumulator into denary.

.....[1]

MARK SCHEME

4(a)(iii)	-37	1
-----------	-----	---

QUESTION

(iii) The ASCII code for 'A' is 41 in hexadecimal.

Calculate the ASCII code in hexadecimal for 'Z'. Show your working.

Working

.....

.....

ASCII code in hexadecimal for 'Z' [2]

MARK SCHEME

4(b)(iii)	1 mark for correct working, 1 mark for correct answer Working: Code for Z = Code for A + 25 ₁₀ Code for Z = 41 ₁₆ + 25 ₁₀ Code for Z = 41 ₁₆ + 19 ₁₆ Code for Z = 5A ₁₆ Answer: 5A ₁₆	2
-----------	--	----------

QUESTION

(c) X is a register. The current contents of X are:

1	0	0	0	0	1	1	1
---	---	---	---	---	---	---	---

(i) The current contents of register X represent an unsigned binary integer.

Convert the value in X into denary.

..... [1]

MARK SCHEME

8(c)(i)	135	1
---------	-----	----------

9608_s18_qp_11_sc_8.c.ii

QUESTION

(ii) The current contents of register X represent a Binary Coded Decimal.

Convert the value in X into denary.

.....[1]

MARK SCHEME

8(c)(ii)	87	1
----------	----	---

9608_s18_qp_11_sc_8.c.iii

QUESTION

(iii) The current contents of register X stores a two's complement binary integer.

Convert the value in X into denary.

.....[1]

MARK SCHEME

8(c)(iii)	−121	1
-----------	------	---

9608_w17_qp_13_sc_1.a.i

QUESTION

1 (a) Each of the following bytes represents an integer in two's complement form.

State the denary value.

(i) 0111 0111 Denary [1]

MARK SCHEME

1(a)(i)	119	1
---------	-----	---

9608_w17_qp_13_sc_1.a.ii

QUESTION

(ii) 1000 1000

Denary

[1]

MARK SCHEME

1(a)(ii)

–120

1

9608_w17_qp_13_sc_1.a.iii

QUESTION

(iii) Express the following integer in two's complement form.

–17

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

1(a)(iii)

1	1	1	0	1	1	1	1
---	---	---	---	---	---	---	---

1

9608_w17_qp_13_sc_1.b.i

QUESTION

(b) (i) Convert the following denary integer into Binary Coded Decimal (BCD).

653

.....[1]

MARK SCHEME

1(b)(i)

0110 0101 0011

1

QUESTION

The following shows two of the program instructions in machine code.

Instruction	Machine code	
	Binary	Hexadecimal
OUTCH	1100 0111	C7
JNE CarryOn	A	B

Each of the numbers **A** and **B** represents the complete instruction in two bytes, one byte for the op code and one byte for the operand.

(ii) Use the following instruction set to write the numbers for **A** and **B**.

A (binary)
B (hexadecimal)
[3]

MARK SCHEME

4(d)(ii)	A – 1110 0110 0110 1000 (1) (1) B – E6 68 (1)	3
----------	---	----------

QUESTION

(d) Computer scientists often write binary representations in hexadecimal.

(i) Write the hexadecimal representation for this instruction:

0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

.....[2]

MARK SCHEME

4(d)(i)	07 C2 07 C2	1 1 2
---------	-------------------	--------------------

9608_s17_qp_13_sc_4.d.ii

QUESTION

(ii) A second instruction has been written in hexadecimal as:

05 3F

Write the equivalent assembly language instruction, with the operand in denary.

.....[2]

MARK SCHEME

4(d)(ii)	LDI 63		2
	LDI 63	1 1	

9608_s17_qp_12_sc_5.c.i

QUESTION

(c) Computer scientists often write binary representations in hexadecimal.

(i) Write the hexadecimal representation for the following instruction.

0

0

0

1

0

1

0

0

0

1

0

1

1

1

1

0

.....[2]

MARK SCHEME

5(c)(i)	14 5E		2
	14 5E	1 1	

9608_s17_qp_12_sc_5.c.ii

QUESTION

(ii) A second instruction has been written in hexadecimal as:

16 4D

Write the assembly language for this instruction with the operand in denary.

.....[2]

MARK SCHEME

5(c)(ii)	LDR #77		2
	LDR #77	1 1	

9608_s17_qp_11_sc_4.d.i

QUESTION

(d) Computer scientists often write binary representations in hexadecimal.

(i) Write the hexadecimal representation for this instruction:

00000111

11000010

.....[2]

MARK SCHEME

4(d)(i)	07 C2		2
	07 C2	1 1	

9608_s17_qp_11_sc_4.d.ii

QUESTION

(ii) A second instruction has been written in hexadecimal as:

05 3F

Write the equivalent assembly language instruction, with the operand in denary.

.....[2]

MARK SCHEME

4(d)(ii)	LDI 63		2
	LDI 63	1 1	

9608_w16_qp_13_sc_3.a.i

QUESTION

3 (a) (i) Convert the denary number 46 to an 8-bit binary integer.

.....
.....[1]

MARK SCHEME

3 (a) (i) 0 0 1 0 1 1 1 0 [1]

9608_w16_qp_13_sc_3.a.ii

QUESTION

(ii) Convert the denary integer –46 to an 8-bit two's complement form.

.....
.....[1]

MARK SCHEME

(ii) 1 1 0 1 0 0 1 0 [1]

9608_w16_qp_13_sc_3.a.iii

QUESTION

(iii) Convert the denary number 46 into hexadecimal.

.....
.....[1]

MARK SCHEME

(iii) 2 E [1]

QUESTION

(b) Binary Coded Decimal (BCD) is another way of representing numbers.

- (i) Describe how denary integers larger than 9 can be converted into BCD.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(b) (i) **One mark** for the explanation and **one mark** for the example

- Each denary digit is written as a 4-bit binary number
- Example: 46 = 0100 0110

[2]

QUESTION

- (ii) Describe how an 8-bit BCD representation can be converted into a denary integer.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(ii) **One mark** for the explanation and **one mark** for the example

- Binary number is split up into groups of 4 bits (starting from the right)
// Each group of 4 bits is converted to a denary digit
- Example: 0011 0111 = 37

[2]

QUESTION

(ii) Show the contents of the Accumulator after execution of the instruction:

LDX 800

Index Register:

0	0	0	0	1	0	0	1
---	---	---	---	---	---	---	---

Accumulator:

--	--	--	--	--	--	--	--

Explain how you arrived at your answer.

.....

[3]

MARK SCHEME

(ii) **One mark** for answer and **two marks** for explanation

Accumulator:

1	1	0	0	0	0	1	0
---	---	---	---	---	---	---	---

- Index Register contains $1001 = 9$
- $800 + 9 = 809$

[3]

QUESTION

- 4 Hexadecimal, Binary Coded Decimal (BCD) and binary values are shown below.

Draw a line to link each value to its correct denary value.

Hexadecimal, BCD, binary**Denary**

Hexadecimal:

3A

93

-65

BCD representation:

0100 1001

58

-63

Binary integer:

01011101

73

Two's complement
binary integer:

11000001

49

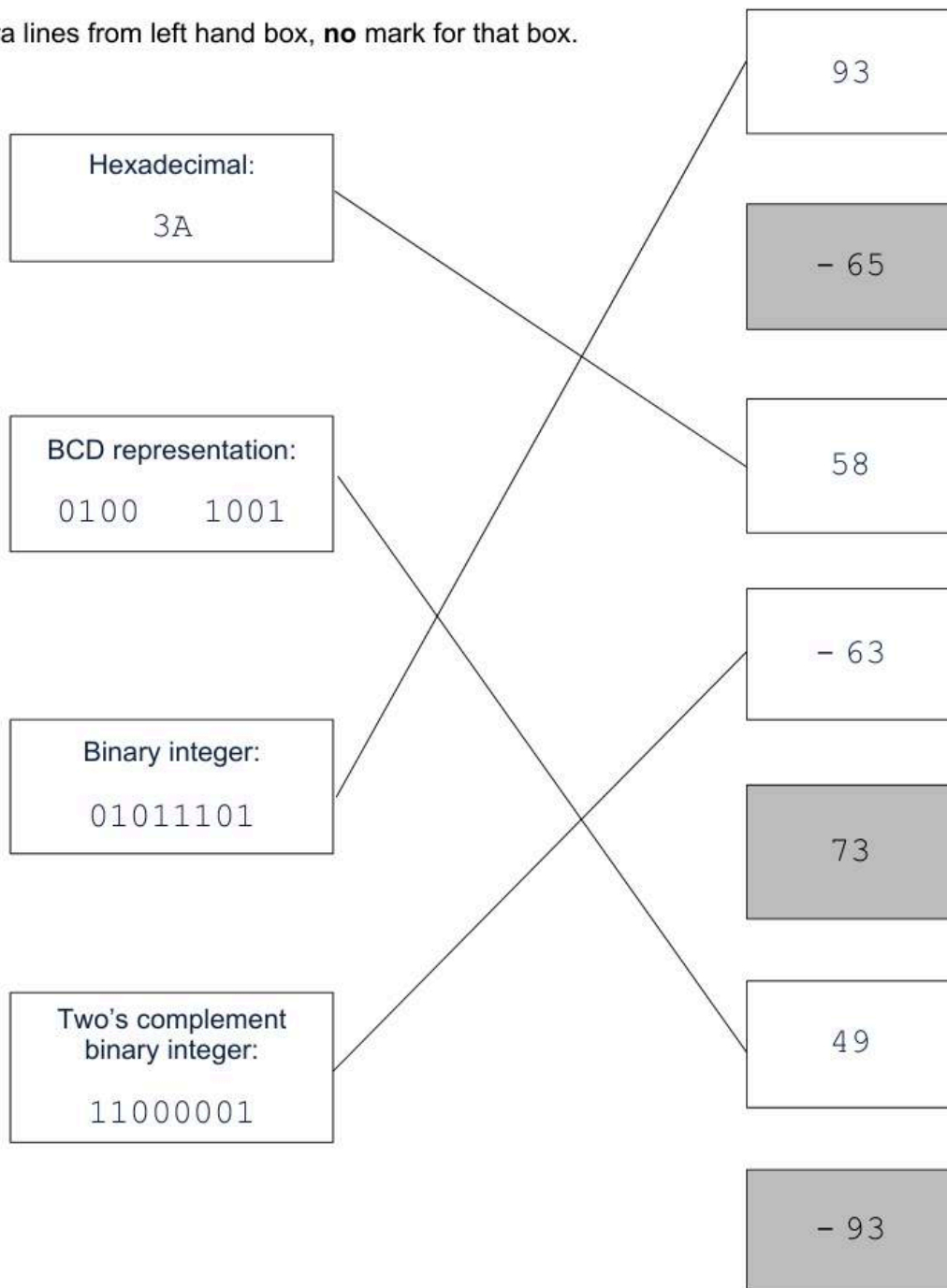
-93

[4]

MARK SCHEME

4 **ONE** mark for each correct line.

Extra lines from left hand box, **no** mark for that box.



[4]

9608_w16_qp_11_sc_3.a.i

QUESTION

3 (a) (i) Convert the denary number 46 to an 8-bit binary integer.

.....
.....[1]

MARK SCHEME

3 (a) (i) 0 0 1 0 1 1 1 0 [1]

9608_w16_qp_11_sc_3.a.ii

QUESTION

(ii) Convert the denary integer –46 to an 8-bit two's complement form.

.....
.....[1]

MARK SCHEME

(ii) 1 1 0 1 0 0 1 0 [1]

9608_w16_qp_11_sc_3.a.iii

QUESTION

(iii) Convert the denary number 46 into hexadecimal.

.....
.....[1]

MARK SCHEME

(iii) 2 E [1]

QUESTION

(b) Binary Coded Decimal (BCD) is another way of representing numbers.

- (i) Describe how denary integers larger than 9 can be converted into BCD.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(b) (i) **One mark** for the explanation and **one mark** for the example

- Each denary digit is written as a 4-bit binary number
- Example: 46 = 0100 0110

[2]

QUESTION

- (ii) Describe how an 8-bit BCD representation can be converted into a denary integer.
Give an example in your answer.

.....

.....

.....

.....[2]

MARK SCHEME

(ii) **One mark** for the explanation and **one mark** for the example

- Binary number is split up into groups of 4 bits (starting from the right)
// Each group of 4 bits is converted to a denary digit
- Example: 0011 0111 = 37

[2]

9608_s16_qp_13_sc_2.a

QUESTION

2 (a) Convert the following 8-bit binary integer into denary.

01001101

.....[1]

MARK SCHEME

2 (a) 77

[1]

9608_s16_qp_13_sc_2.b

QUESTION

(b) Convert the following denary number into Binary Coded Decimal (BCD).

82

.....[1]

MARK SCHEME

(b) 1000 0010

[1]

9608_s16_qp_13_sc_2.c

QUESTION

(c) Convert the following two's complement integer number into denary.

11001011

.....

.....

.....[2]

MARK SCHEME

(c) -53

[2]

One mark for '53' and one mark for '-'

9608_s16_qp_13_sc_2.d

QUESTION

(d) Convert the following denary number into hexadecimal. Show your working.

198

.....[2]

MARK SCHEME

(d) C6

[2]

One mark for the answer, one mark for the method

- Working e.g. $198 / 16 = 12$, $198 - (12 \times 16) = 6$

9608_s16_qp_12_sc_2.a

QUESTION

2 (a) Convert the following denary integer into 8-bit binary.

55

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

2 (a) 00110111

[1]

9608_s16_qp_12_sc_2.b

QUESTION

(b) Convert the following Binary Coded Decimal (BCD) number into denary.

10000011

.....[1]

MARK SCHEME

(b) 83

[1]

9608_s16_qp_12_sc_2.c

QUESTION

(c) Convert the following denary integer into 8-bit two's complement.

-102

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(c) 10011010

[2]

Marks allocated as follows:

1 mark for the most significant bit

1 mark for the remaining 7 bits

9608_s16_qp_12_sc_2.d

QUESTION

(d) Convert the following hexadecimal number into denary.

4E

.....[1]

MARK SCHEME

(d) 78 [1]

9608_s16_qp_11_sc_2.a

QUESTION

2 (a) Convert the following denary integer into 8-bit binary.

55

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

2 (a) 00110111 [1]

9608_s16_qp_11_sc_2.b

QUESTION

(b) Convert the following Binary Coded Decimal (BCD) number into denary.

10000011

.....[1]

MARK SCHEME

(b) 83 [1]

9608_s16_qp_11_sc_2.c

QUESTION

(c) Convert the following denary integer into 8-bit two's complement.

-102

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(c) 10011010

[2]

Marks allocated as follows:

1 mark for the most significant bit

1 mark for the remaining 7 bits

9608_s16_qp_11_sc_2.d

QUESTION

(d) Convert the following hexadecimal number into denary.

4E

.....[1]

MARK SCHEME

(d) 78

[1]

9608_s15_qp_13_sc_1.a.i

QUESTION

1 (a) (i) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

124

--	--	--	--	--	--	--	--

−77

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

1 (a) (i)

124

0	1	1	1	1	1	0	0
---	---	---	---	---	---	---	---

−77

1	0	1	1	0	0	1	1
---	---	---	---	---	---	---	---

[2]

9608_s15_qp_13_sc_1.a.ii

QUESTION

(ii) Convert the two numbers in **part (a) (i)** into hexadecimal.

124

-77 [2]

MARK SCHEME

(ii) 124: 7 C

-77: B 3

[2]

9608_s15_qp_13_sc_1.b.i

QUESTION

(b) Binary Coded Decimal (BCD) is another way of representing numbers.

(i) Write the number 359 in BCD form.

..... [1]

MARK SCHEME

(b) (i) 0 0 1 1 0 1 0 1 1 0 0 1

[1]

9608_s15_qp_12_sc_1.i

QUESTION

1 (i) Convert the following binary number into hexadecimal.

1 0 1 1 1 0 0 0

.....[1]

MARK SCHEME

1 (i) B 8

[1]

9608_s15_qp_12_sc_1.ii

QUESTION

(ii) Convert the following denary number into BCD format.

97

.....[1]

MARK SCHEME

(ii) 1 0 0 1 0 1 1 1

[1]

9608_s15_qp_12_sc_1.iii

QUESTION

(iii) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

114

--	--	--	--	--	--	--	--

-93

--	--	--	--	--	--	--	--

[2]

MARK SCHEME

(iii)

114

0	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---

-93

1	0	1	0	0	0	1	1
---	---	---	---	---	---	---	---

[2]

9608_s15_qp_11_sc_1.i

QUESTION

1

(i) Convert the following binary number into hexadecimal.

1 0 1 1 1 0 0 0

.....[1]

MARK SCHEME

1

(i) B 8

[1]

9608_s15_qp_11_sc_1.ii

QUESTION

(ii) Convert the following denary number into BCD format.

9 7

.....[1]

MARK SCHEME

(ii) 1 0 0 1 0 1 1 1

[1]

9608_s15_qp_11_sc_1.iii

QUESTION

(iii) Using two's complement, show how the following denary numbers could be stored in an 8-bit register:

114

- 93

[2]

MARK SCHEME

(iii)

114

0	1	1	1	0	0	1	0
---	---	---	---	---	---	---	---

– 93

1	0	1	0	0	0	1	1
---	---	---	---	---	---	---	---

[2]

Perform binary addition and subtraction

9608_w18_qp_13_sc_2.b.iv

QUESTION

(iv) Show the result on the general purpose register (X) after the following instruction is run.

INC X

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

2(b)(iv)	<table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	1	1	1	1	0	0	1	1	1
1	1	1	1	0	0	1	1			

QUESTION

- 4 The table shows assembly language instructions for a processor which has one general purpose register, the Accumulator (ACC) and an index register (IX).

Instruction		Explanation
Op code	Operand	
LDD	<address>	Direct addressing. Load the contents of the given address to ACC.
LDX	<address>	Indexed addressing. Form the address from <address> + the contents of the index register. Copy the contents of this calculated address to ACC.
STO	<address>	Store contents of ACC at the given address.
ADD	<address>	Add the contents of the given address to ACC.
INC	<register>	Add 1 to the contents of the register (ACC or IX).
DEC	<register>	Subtract 1 from the contents of the register (ACC or IX).
CMP	<address>	Compare contents of ACC with contents of <address>.
JPE	<address>	Following a compare instruction, jump to <address> if the compare was True.
JPN	<address>	Following a compare instruction, jump to <address> if the compare was False.
JMP	<address>	Jump to the given address.
OUT		Output to screen the character whose ASCII value is stored in ACC.
END		Return control to the operating system.

The diagram shows the contents of the index register:

Index register:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

- (a) Show the contents of the index register after the execution of the instruction:

INC IX

Index register:

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

4 (a) 11001110

[1]

Describe practical applications where Binary Coded Decimal (BCD) and Hexadecimal are used

9608_w19_qp_11_sc_5.c.ii

QUESTION

(ii) Identify **one** practical application where BCD is used.

.....
..... [1]

MARK SCHEME

5(c)(ii)	Any scenario where a single digit needs to be transmitted / displayed e.g. calculator / digital clock	1
----------	--	----------

9608_w17_qp_13_sc_1.b.iii

QUESTION

(iii) Describe a practical application where BCD is used.

.....
.....[1]

MARK SCHEME

1(b)(iii)	A string of digits on any electronic device displaying numeric values	1
-----------	---	----------

9608_s15_qp_13_sc_1.b.ii

QUESTION

(ii) Describe a use of BCD number representation.

.....
..... [2]

MARK SCHEME

- (ii) • when denary numbers need to be electronically coded
• e.g. to operate displays on a calculator where each digit is represented
• decimal fractions can be accurately represented

[2]

Show understanding of and be able to represent character data in its internal binary form, depending on the character set used

9608_s20_qp_11_sc_4.c.i

QUESTION

- (c) The digital camera allows a user to add text to an image. The text is encoded as ASCII values.

The table shows the ASCII denary values for five characters.

Character	ASCII denary value
a	97
b	98
c	99
d	100
e	101

- (i) Give the 8-bit binary value for the ASCII character 'b'.

.....
..... [1]

MARK SCHEME

4(c)(i)	0110 0010	1
---------	-----------	---

9608_s20_qp_11_sc_4.c.ii

QUESTION

- (ii) Complete the table by writing the ASCII denary value for the character 't' **and** its hexadecimal equivalent.

Character	t
ASCII denary value	
Hexadecimal value	

[2]

MARK SCHEME

4(c)(ii)	1 mark for each correct line	2						
	<table><tr><td>Character</td><td>t</td></tr><tr><td>ASCII denary value</td><td>116</td></tr><tr><td>Hexadecimal value</td><td>74</td></tr></table>	Character	t	ASCII denary value	116	Hexadecimal value	74	
Character	t							
ASCII denary value	116							
Hexadecimal value	74							

9608_w19_qp_11_sc_5.d

QUESTION

(d) One example of a character set used by computers is ASCII.

Describe how one character is represented in a character set.

.....

 [2]

MARK SCHEME

5(d)	1 mark per bullet to max 2 Each character is represented by a unique ... denary / hexadecimal / binary number	2
------	--	----------

9608_s19_qp_12_sc_3.d.i

QUESTION

(d) The ASCII character code for 'A' is 65 in denary.

(i) Convert the denary ASCII character code for 'A' into 8-bit binary.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

3(d)(i)	1 mark for correct answer 0100 0001	1
---------	---	----------

9608_s19_qp_12_sc_3.d.ii

QUESTION

(ii) Convert the denary ASCII character code for 'A' into hexadecimal.

..... [1]

MARK SCHEME

3(d)(ii)	1 mark for correct answer 41	1
----------	--	----------

9608_s19_qp_12_sc_3.d.iii

QUESTION

(iii) The Unicode character code for 'G' is 0047 in hexadecimal.

State, in hexadecimal, the Unicode character code for 'D'.

..... [1]

MARK SCHEME

3(d)(iii)	1 mark for correct answer 0044	1
-----------	--	----------

9608_s18_qp_12_sc_4.b.i

QUESTION

(b) The binary integer represents a character from the computer's character set.

(i) Define the term **character set**.

.....
.....[1]

MARK SCHEME

4(b)(i)	1 mark from: The <u>symbols</u> that the <u>computer</u> recognises/uses A list of <u>characters</u> recognised by the <u>computer</u> hardware and software	1
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9608_s18_qp_12_sc_4.b.ii

QUESTION

- (ii) Explain the differences between the **ASCII** and **Unicode** character sets.

.....

[2]

MARK SCHEME

4(b)(ii)	1 mark per bullet to max 2 UNICODE has greater range of characters than ASCII UNICODE represents most written languages in the world while ASCII does not ASCII used for English only ASCII uses 7 or 8 bits or one byte whereas UNICODE uses up to 4 bytes per character UNICODE is standardised while ASCII is not	2
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9608_s18_qp_12_sc_4.b.iii

QUESTION

- (iii) The ASCII code for 'A' is 41 in hexadecimal.

Calculate the ASCII code in hexadecimal for 'Z'. Show your working.

Working

.....

ASCII code in hexadecimal for 'Z'[2]

MARK SCHEME

4(b)(iii)	1 mark for correct working, 1 mark for correct answer Working: $\text{Code for Z} = \text{Code for A} + 25_{10}$ $\text{Code for Z} = 41_{16} + 25_{10}$ $\text{Code for Z} = 41_{16} + 19_{16}$ $\text{Code for Z} = 5A_{16}$ Answer: $5A_{16}$	2
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QUESTION

(c) The program used the ASCII coding system for character codes. An alternative coding system is Unicode.

(i) Give **two** disadvantages of using ASCII code.

- 1
-
- 2
-[2]

MARK SCHEME

(c) (i) Any **two** from:

- Only 128 / 256 characters can be represented
- Uses values 0 to 127 (or 255 if extended form) / one byte
- Many characters used in other languages cannot be represented
- In extended ASCII the characters from 128 to 255 may be coded differently in different systems

[2]

QUESTION

(ii) Describe how Unicode is designed to overcome the disadvantages of ASCII.

-
-
-
-[2]

MARK SCHEME

(ii) Any **two** from:

- Uses 16, 24 or 32 bits / two, three or four bytes
- Unicode is designed to be a superset of ASCII
- Designed so that most characters (in other languages) can be represented

[2]

QUESTION

(c) The program used the ASCII coding system for character codes. An alternative coding system is Unicode.

(i) Give **two** disadvantages of using ASCII code.

- 1
-
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-[2]

MARK SCHEME

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-
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-
-[2]

MARK SCHEME

(ii) Any **two** from:

- Uses 16, 24 or 32 bits / two, three or four bytes
- Unicode is designed to be a superset of ASCII
- Designed so that most characters (in other languages) can be represented

[2]

QUESTION

(b) Complete the trace table on the opposite page for the following assembly language program.

50	LDD 100
51	ADD 102
52	STO 103
53	LDX 100
54	ADD 100
55	CMP 101
56	JPE 58
57	JPN 59
58	OUT
59	INC IX
60	LDX 98
61	ADD 101
62	OUT
63	END
...	
100	20
101	100
102	1
103	0

IX (Index Register)

Selected values from the ASCII character set:

ASCII Code	118	119	120	121	122	123	124	125
Character	v	w	x	y	z	{		}

Trace table:

Instruction address	Working space	ACC	Memory address				IX	OUTPUT
			100	101	102	103		
			20	100	1	0	1	
50								
51								
52								
53								
54								
55								

[7]

MARK SCHEME

(b)

Instruction address	Working space	ACC	Memory address				IX	OUTPUT
			100	101	102	103		
			20	100	1	0	1	
50		20						
51		21						
52						21		
53		100						
54		120						
55								
56								
57								
59							2	
60		20						
61		120						
62								'x'
63								

One mark for each shaded block.

[7]

- Contents of the Accumulator in first 2 lines (instruction addresses 50 and 51)
- Updating address 103 (instruction 52)
- Loading the Accumulator and addition (instructions 53 and 54)
- Not executing instruction 58
- Incrementing the index register (instruction 59)
- Loading the Accumulator and addition (instructions 60 and 61)
- Correct output of 'x' (instruction 62)

QUESTION

(b) Complete the trace table on the opposite page for the following assembly language program.

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[7]

MARK SCHEME

(b)

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